



Enhanced antifouling strategy with a strong synergistic effect of fluorescent antifouling and contact bacteriostasis using 7-amino-4-methylcoumarin

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ABSTRACT

Low surface energy coating can protect marine equipment from biofouling, however, remaining a challenge in static applications due to the lack of shear force provided by water scour. Herein, we developed a polyurethane with polydimethylsiloxane (PDMS), polytetrahydrofuran (PTMG) as main chains and 7-amino-4-methylcoumarin as antifouling group by condensation polymerization. Besides the fluorescence response to the ultraviolet band in seawater with high transmittance, the typical benzopyrone framework of 7-amino-4-methylcoumarin endows antibacterial properties. Owing to the strong synergistic effect of fluorescence antifouling and contact bacteriostasis mechanism, the PDMS-based fouling-release surface achieves a superior antifouling effect with an extremely small amount (less than 0.20 wt%) of the antifouling group, which is almost the lowest among previous research. And 7-amino-4-methylcoumarin can be naturally derived from plants, which explored novel research for environmentally friendly antifouling strategy.

1. Introduction

The marine antifouling coating has been recognized as one of the optimal solutions towards fouling-resistant.[1–3] At present, the mainstream antifouling coating in the market is mostly accompanied by the release of biocides or toxic materials.[4] With the development of marine economy and the increasing attention of countries to marine ecological environment protection, the low surface energy fouling-release coating without biocides meets the requirements of environmental protection, and there is no such issue of the decline of antifouling effect due to the biocides releasing. Polydimethylsiloxane (PDMS)-based fouling-release coating enjoys the properties of low surface energy and low modulus of elasticity, making it an ideal marine antifouling material.[5,6]

The surface chemistry, physical properties, durability, and adhesion scheme of low surface energy coating are very important for the practical fouling-release effect of the coating in the marine environment, especially, there remains a major bottleneck for fouling release coating in static applications, namely, offshore drilling, exploration equipment, wharf, aquaculture cage, and other artificial facilities surface, where lack of shear force provided by water scour to release the fouling.[7,8]

To realize excellent adhesion with the substrate while maintaining low surface energy characteristics, multiple soft segments with different polarities were taken as soft segments.[9,10] Limited antifouling under static conditions endows the growth of fouling like diatoms and bacteria, so functional groups containing hydrophilic, amphiphilic compounds or zwitterions were introduced to resist protein and primary organic fouling.[6,11–14] The reactive antifouling group can be introduced into the resin for long-term antifouling.[3,6,15] The degradation of coating properties and the environmental hazard caused by the high addition of antifouling groups are insurmountable problems, which seriously limit their large-scale applications.[16–18]

In recent years, enormous studies have been carried out on the antimicrobial substances extracted from plants, one of which is coumarin compounds, which have biological activities thanks to the typical benzopyrone framework, such as anti-tumor, anti-bacterial, anti-virus, absorption of ultraviolet rays and anticoagulant, etc.[19–23] As natural antibacterial agents, they exert an inhibitory effect on the growth of microorganisms and the secretion of extracellular polymeric substances (EPS) and metabolites in the early stage of fouling.[24]

Among many coumarin compounds, 7-amino-4-methylcoumarin not only has antibacterial biological activity but also exhibits ultraviolet

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(UV) fluorescence response with maximum emission wavelength of about 480 nm.[25] Very coincidentally, seawater is highly transparent for UV radiation.[26] Research proved that in the mid-latitude eastern Mediterranean Sea shows that for every 1 m drop in seawater depth, the intensity of UV at 310 nm decreases by 14%; the intensity of UV at 375 nm reduces by 5%; for visible light (blue light) at 465 nm, the corresponding value is 3%, and in highly transparent seawater, UV radiation can reach as far down as 20 m, which make photochemical processes possible to proceed.[27]

In coatings, the introduction of photo-response materials can effectively reduce the deposition of diatoms on the coating surface to prevent and control marine biological pollution in the primary stage.[28–30] The mechanism of fluorescent antifouling is based on the interrelation of physiological activities of cells and illumination, which influences spatial interference between chloroplasts and holdfast-like structures inside the thecae.[28] And, with additional light, diatoms will perform photosynthesis with higher priority rather than settlement.[26,31] So the light condition, for example, light wavelength and light intensity, can be a potential approach to inhibit primary biofouling like diatom.[32] In some fluorescent antifouling research, silicone elastomer with colored (afterglow) phosphor powder indicated the inhibition of diatom initial settlement with the fouling-release characteristics.[29,30] Some photocatalysts, such as TiO_2 , were also applied as photo-response material, with the antifouling mechanism of reactive oxygen species (ROS) sterilization.[33–35] All previous studies entail a strong light supply to irradiate the photo-response process, and in the environment of no light or weak light, the efficiency of antifouling will decrease, which limits the practical application in the marine environment.

Herein, we explored a novel enhanced antifouling strategy for fouling-release coating integrated with fluorescent antifouling and contact bacteriostasis mechanisms. The resulting PDMS-based polyurethane with 7-amino-4-methylcoumarin was synthesized by a simple condensation polymerization method, possessing a strong synergistic effect of two antifouling mechanisms, exhibited significant antifouling performance under all-weather conditions of light, with an extremely low addition (less than 0.20 wt%, Table S1) of antifouling groups.

2. Results and discussion

2.1. Fabrication of PDMS-based polyurethane with 7-amino-4-methylcoumarin.

The PDMS-based polyurethane (PDMS-x-PU-y) was synthesized via condensation polymerization and modified with 7-amino-4-methylcoumarin and terminated by KH-550 (Fig. 1B). As shown in Fig. 1C, the chemical structure and surface elemental compositions of PDMS-x-PU-y were identified by analyzing Fourier transform infrared (FTIR) and energy-dispersive X-ray spectrometer (EDS). As shown in Fig. S1, the peak at 3347 cm^{-1} was assigned to the stretching vibration of the -N-H bond in the urethane segments (-NH-CO-O).[36] The synthesis of the PDMS-based polyurethane was confirmed by the decrease and disappearance of the characteristic -NCO peak at 2270 cm^{-1} , which indicate that the NCO groups were reacted with the amino group of 7-amino-4-methylcoumarin and terminated by KH-550. More details about the ^1H NMR spectra can be found in Fig. S2.

As demonstrated in Fig. 1C, the peaks in the range of $2850\text{--}2935\text{ cm}^{-1}$ correspond to the -CH stretching vibrations (asymmetric and symmetric vibrations) of the alkyl chain present in PTMG and HT-PDMS. The characteristic absorption of C=O (1735 cm^{-1}) stretching vibrations were clearly visible. The peak at 1259 cm^{-1} represents the in-plane bending vibration peak of Si-CH₃, the coordination peak at 1009 cm^{-1} represents stretching of Si-O in Si-O-Si groups, and the peak at 864 cm^{-1} represents the stretching vibration peak of Si-C. There is also Si-O-Si asymmetric stretching mode at 1100 cm^{-1} and Si-O-Si symmetric stretching vibration at 1120 cm^{-1} .^[37] More importantly, with the constant mole of HT-PDMS, these specific peaks of the samples are more obvious with the increase of the molecular weight of HT-PDMS. The prepared PDMS-based polyurethane could directly crosslink by the moisture-curing process at room temperature.

2.2. Characteristics analysis of the PDMS-based polyurethane coatings.

The chemical composition of the PDMS-based polyurethane (PDMS-x-PU-0.03) can be inferred by analyzing the EDS spectra of the samples (Fig. S4). In Fig. 1C, there is a higher ratio of silicon to nitrogen on the

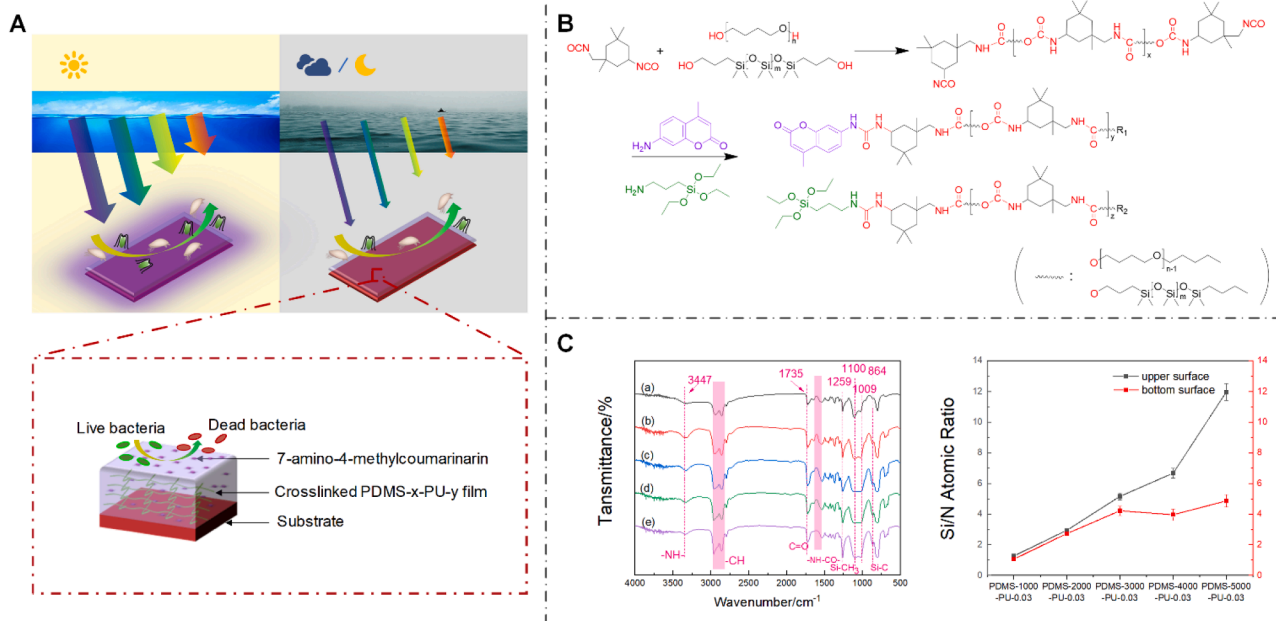


Fig. 1. (A) Schematic illustration of the synergistic effect of fluorescent antifouling and contact bacteriostasis. (B) Synthesis of PDMS-x-PU-y with 7-amino-4-methylcoumarin (C) FT-IR spectra and Si/N atomic ratio^a of upper surface and bottom surface of PDMS-x-PU-0.03^a The actual values were determined based on the EDS measurement.

upper surface of the coatings in the EDS data than the bottom surface of the coatings. What's more, with the increase of the molecular weight of HT-PDMS, the tendency of self-stratifying due to element rearrangement becomes more prominent. The self-stratifying effect confirmed that the silicone segments providing weak interaction with the substrate migrate to the upper surface and endows the hydrophobicity and low surface energy of the coating. In this way, the complex process in the application of low surface energy coating can be reduced.

The surface morphologies of the coating samples were detected by FESEM and AFM. Fig. 2a, b, and c show the FESEM images of the PDMS-1000-PU-0.03, PDMS-3000-PU-0.03, PDMS-5000-PU-0.03, the coating has good flatness and low roughness on macro and micro nanoscale. Results revealed that under the same magnification, the coating surface containing PDMS with a molecular weight of 1000 formed more significant phase separation. Keeping the proportion of soft and hard segments unchanged, when using higher molecular weight PDMS, the surface roughness decreased accordingly.

AFM was applied to further confirm the surface morphology of the coating samples at the microscale (Fig. 2d, e, and f). The surface exhibited discontinuous regions with bright and dark domains due to different modulus. For the harder or rigid materials, the phase angle is small, while for softer material the phase angle is higher, such as PDMS. [38] Hence, the bright phase indicated the PDMS domains, and the PDMS domains appeared on the coating surface in the form of the bulge. [39] As for PDMS-1000-PU-0.03, alternating soft and hard regions indicated apparent phase separation. Since the introduction of large molecular weight PDMS reduces the relative density of hydrogen bond between urethane N-H and ether bonds, the degree of phase separation decreases. The reduction in phase separation combined with more silicon elements leads to a reduction in the roughness of the coating.

The coatings have high transparency (transmittance more than 98.5%) in the range of visible light region (475–800 nm) as shown in Fig. 3A, and have obvious absorption in the ultraviolet band especially in the range of 300–420 nm, which is consistent with the high-efficiency band of sunlight projecting seawater. [27] The coating has an obvious fluorescence response to 365 nm ultraviolet light. On the other hand, this coating may also have the potential to be adapted widely to the fields of solar cell transparent plate and ocean periscope, etc.

2.3. Surface energy and wettability.

Little amount of 7-amino-methylcoumarin did not significantly affect

the surface energy of the coating (Table S2). Fig. 3B shows the values of the surface energy of PDMS-x-PU-0.03 coating range from 21 to 22 mN/m due to the enrichment of the PDMS segment providing low surface energy and contributing dispersive component (Table S2). According to the relationship between surface energy and fouling release curve, known as Baier curve, the surface energy corresponds to optimal fouling release performance is between 20 and 25 mN/m. [40,41] The low surface energy properties of the PDMS-based polyurethane coatings agreed with the range that helps to remove settled organisms. The water contact angles (WCAs) of all samples were above 105°, possessing excellent hydrophobicity (Fig. 3C).

2.4. Adhesion and lap shear tensile tests.

The organisms are supposed to be removed by the shear forces generated by ship movement or mild mechanical cleaning devices, so the mechanical durability of prepared PDMS-x-PU-y coatings was conducted by pull-off adhesion tests and lap shear tests.

The adhesion of the antifouling coating to substrate based on low surface energy is a huge bottleneck in its industrial application. [3,42,43] The surface energy of the coating is supposed to be low enough and in the appropriate range of the Baier curve, while, such low surface energy should not affect the adhesion strength of the coating in application environments. PDMS content will affect the polarity of the coating and then affect its adhesion performance. [15] The mechanical durability of samples of PDMS-x-PU-0.03 were quantified through pull-off adhesion tests (Fig. 3D) which indicated the adhesion strength with the epoxy-coated substrate was enough for the application in marine environments and lap shear tests were also carried out to simulate the shear force during the navigation of the ship (Fig. S3). Coatings with the PDMS molecular weight between 2000 and 4000 displayed higher adhesion strength to epoxy primer substrate. When the molecular weight of PDMS is relatively small, the motion of the hard segments is hindered and the elongation at break is small (Fig. S3). When the soft segment is increased to a certain extent, the elongation at break decreases due to the poor mechanical properties of the soft segment. When the molecular weight of PDMS is about 3000, the coating has excellent shear resistance performance, and the elongation of the coating is larger when damaged, reaching the best range of mechanical properties (Fig. S3).

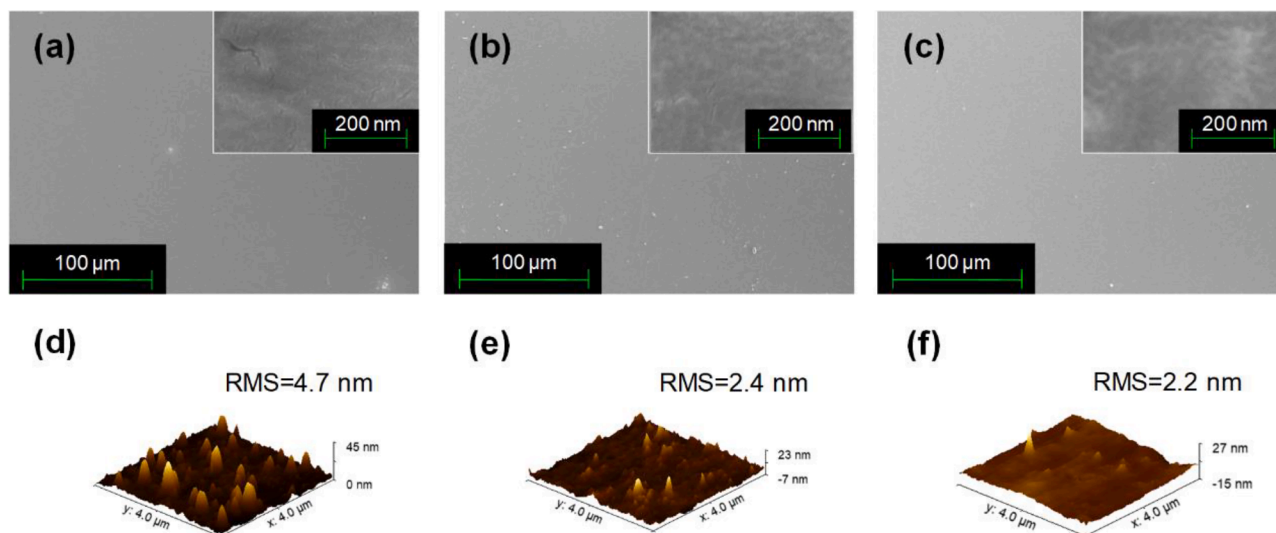


Fig. 2. The FESEM images of (a) PDMS-1000-PU-0.03 (b) PDMS-3000-PU-0.03 (c) PDMS-5000-PU-0.03 and 3D AFM height images of (d) PDMS-1000-PU-0.03 (e) PDMS-3000-PU-0.03 (f) PDMS-5000-PU-0.03.

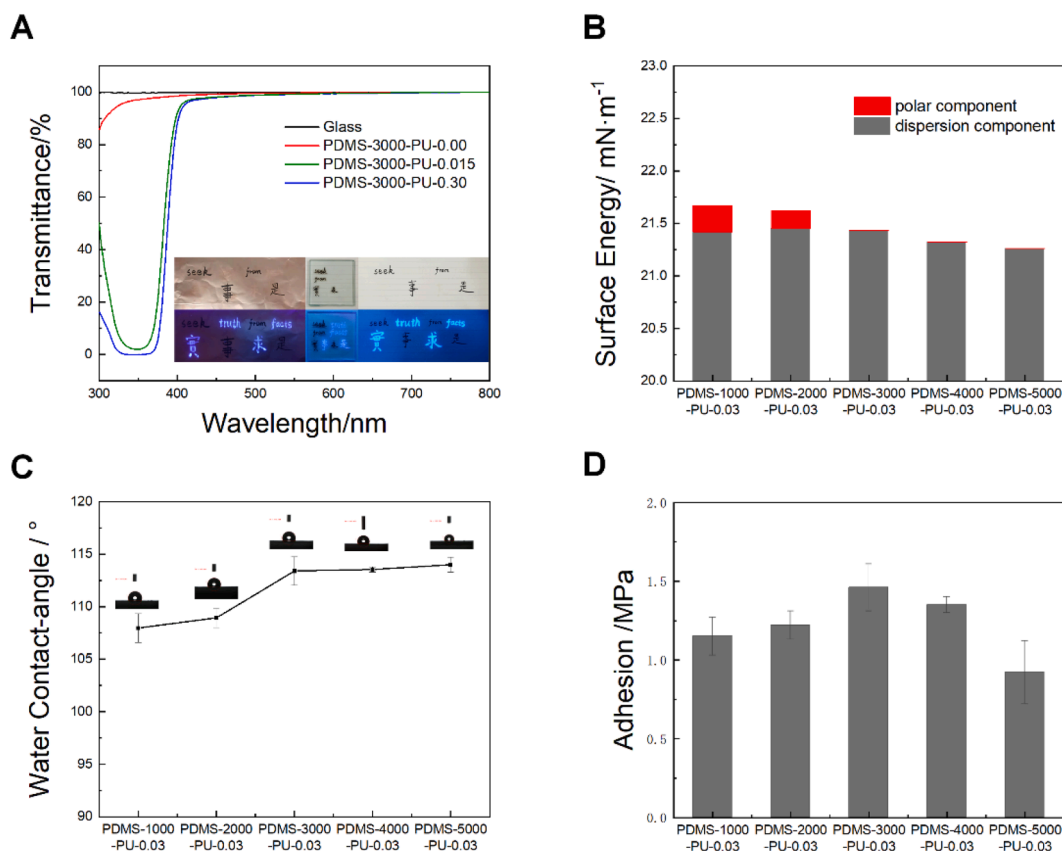


Fig. 3. (A) Transparency of glass, PDMS-3000-PU-0.00, PDMS-3000-PU-0.015, and PDMS-3000-PU-0.015 coating brushed on copper foil, glass, and paper, and corresponding images under 365 nm UV light. (B) Surface energy (C) Water contact-angle (WCA) of the PDMS-x-PU-0.03 coatings. (D) Adhesion strength of coatings with the epoxy-coated substrate by pull-off adhesion tests.

2.5. Self-cleaning, Bacteriostatic, algae plate inoculation Experiments, and marine field antifouling Test.

Self-cleaning Test. The self-cleaning test was carried out for inorganic pollutants with different hydrophilicity. The performance of prepared PDMS-x-PU-y coatings was demonstrated with copper (II) chloride dihydrate ($\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$) particles as a hydrophilic contaminant and carbon particles as hydrophobic contaminant then removed with water droplets. As shown in Fig. 4A, B, the contaminants were put on the surface of the samples with an inclination of about 13°, and the deionized water drops from above the contaminants. The hydrophilic contaminant was dissolved and slide down the coating surface. The most hydrophobic contaminants were washed away by the water droplets, except for very fine carbon toner residues on the coating surface. Therefore, the self-cleaning property of the coating plays a role in the elimination of various solid pollutants in practical applications.

Bacterial Biofouling Resistance Test. As a naturally occurring oxygen-containing heterocycle with a typical benzopyrone framework, 7-amino-4-methylcoumarin has interaction with various enzymes and receptors in organisms through weak bond interactions, which enables 7-amino-4-methylcoumarin-modified PDMS-based polyurethane coatings to exhibit biocidal activity through the contact-killing mechanism. [22,23,44] More specifically, the electron-rich and good charge-transport properties of the π - π conjugated system and latent ability of noncovalent interactions endue coumarin-based derivatives biological activities, which can act as antibacterial agents to destroy the membrane integrity of the bacterial cell.

The antibacterial properties of polyurethane coatings PDMS-3000-PU-y were investigated with coumarin content as a gradient by plate count method. Gram-negative *E. coli* was taken as model bacteria. The bacterial resistance performance can be reflected by the colony numbers

on the LB agar plates (Fig. 4C). Compared with the control sample of the virgin ordinary glass, the existence of hydrophobic low surface energy coating reduced the adhesion of bacteria on the substrate surface to a certain extent. The coatings modified with 7-amino-4-methylcoumarin led to a significant reduction in the *E. coli* colony numbers after 24 h of incubation, and with the increasing 7-amino-4-methylcoumarin content in the polyurethane, the antibacterial effect became more significant. The results of the bacterial biofouling resistance efficiency for different samples are presented in Table 1. And the bactericidal rate against bacteria *E. coli* was above 90% when the amount of 7-amino-4-methylcoumarin in the sample was mere 0.654 wt% (0.015 g, PDMS-3000-PU-0.015), while with the increasing coumarin content to mere 1.307 wt% (0.03 g, PDMS-3000-PU-0.03), the coating exhibited an average antibacterial efficiency of 98.17%. The result of bacterial biofouling resistance shows that the introduction of 7-amino-4-methylcoumarin in the PDMS-based polyurethane coating endows the coating's excellent antimicrobial activity. Even if the coumarin is not added, the settlement of the bacteria will also be reduced due to the low surface energy characteristics of the coating. Based on these data, the coatings in this study can further be applied in biomedical and related fields.

Algae Plate Inoculation Test. In the early stage of biological fouling, diatoms are one of the most important contributors in the formation of macrofouling after the initial stage of fouling. Marine algae (mainly diatom) was used to evaluate the antifouling ability against micro-fouling of the coatings (Fig. 4D). Under the natural light with day and night variation for 7 days, the colonization and growth of diatoms on the coating surface were studied, compared with samples preserved away from light to simulate the all-weather conditions and to prove the fluorescent antifouling performance of the coating. A very small amount of coumarin could significantly improve the antifouling effect. The coatings had better algae resistance performance with sunlight. On the

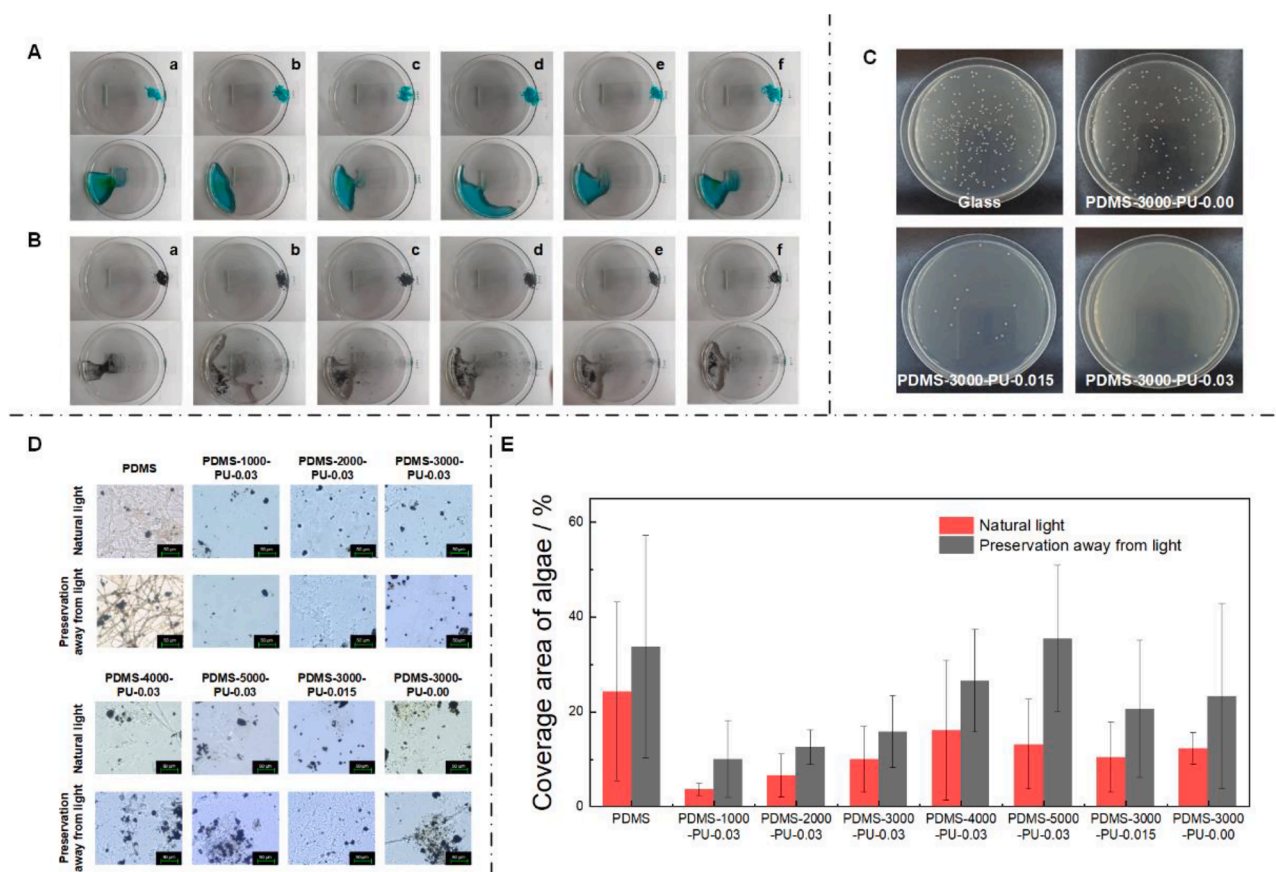


Fig. 4. Self-cleaning test by (A) copper (II) chloride dihydrate ($\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$) particles, and (B) carbon particles. Using continuous water droplets to rinse the contaminated coating surface. Images (a), (b), (c), (d), (e), (f) were taken before (top) and after (bottom) a few drops of water washing the surface of glass, PDMS-1000-PU-0.03, PDMS-2000-PU-0.03, PDMS-3000-PU-0.03, PDMS-4000-PU-0.03, PDMS-5000-PU-0.03, respectively. The inclination of the sample is about 13° (C) Bacterial biofouling resistance test with *E. coli* after 5 h of contact with the sample blank glass plate, PDMS-3000-PU-0.00, PDMS-3000-PU-0.015 and PDMS-3000-PU-0.03 (D) Representative microscope images and (E) Percent of the coverage area of algae on different coatings kept in natural light or preserved away from light after 7 days.

Table 1

Antibacterial Efficiency for the 7-amino-4-methylcoumarin-modified PDMS-based polyurethane coatings (PDMS-3000-PU-y).

Samples	Average E_b / %
Glass	/
PDMS-3000-PU-0.00	60.37
PDMS-3000-PU-0.015	90.49
PDMS-3000-PU-0.03	98.17

one hand, PDMS-5000-PU-0.03 had a similar coverage rate with pure PDMS coating when preserved in the dark, while under the condition of natural light, the degree of algae inoculation of PDMS-5000-PU-0.03 had a more significant decline. On the other hand, there was no distinct difference in the coverage area of algae of PDMS-3000-PU-y without light, which indicated that fluorescent antifouling made a main contribution to the synergistic antifouling effect. Moreover, the performance of anti-diatom was also related to the hydrophilic degree of the surface, the higher the hydrophilic degree, the better the anti-diatom settlement effect. The algae colonized on these samples can be easily removed under 0.1 MPa water washing (Fig. S5).

Marine Field Antifouling Test. Due to good mechanical performances, PDMS-3000-PU-y were selected to carry out the offshore hanging plate test in the Zhoushan sea area of the East China Sea (Fig. 5). During the first 90 days (between April to July 2020), the marine organisms grew slowly in spring and early summer, and the coatings with coumarin were

not fouled significantly, while the coating free of coumarin began to have marine organisms settling on it. When it comes to 120 days, the marine organisms thrived with a higher speed in the biological growth season (July to August 2020) (Fig. S6), which made the antifouling performance of different coatings significant. After immersion in seawater for 120 days, there was still not too much fouling pressure on the coating surface containing coumarin, performing stable antifouling properties under static conditions. The more coumarin contained in the samples, the less biological colonization happened on the surface. After washing, the fouling on the coating surface can be released, and it can be anticipated when the coating is applied under dynamic conditions, for example on ships, the fouling will be released with the shear force provided by water scour more efficiently (Fig. S7). All in all, the results of the marine field hanging plate demonstrate that the coating has good practical application performance in the marine environment.

3. Conclusion

In conclusion, we developed a novel enhanced antifouling strategy with a strong synergistic effect of fluorescent antifouling and contact bacteriostasis, which showed excellent antifouling performance as a fouling-release coating. The PDMS-based polyurethane coating containing 7-amino-4-methylcoumarin as an antifouling functional monomer was prepared by condensation polymerization, with the increase of PDMS molecular weight, the silicon element tended to migrate to the surface of the coating, this self-stratifying effect was conducive to the low surface energy characteristics of the coating surface and the

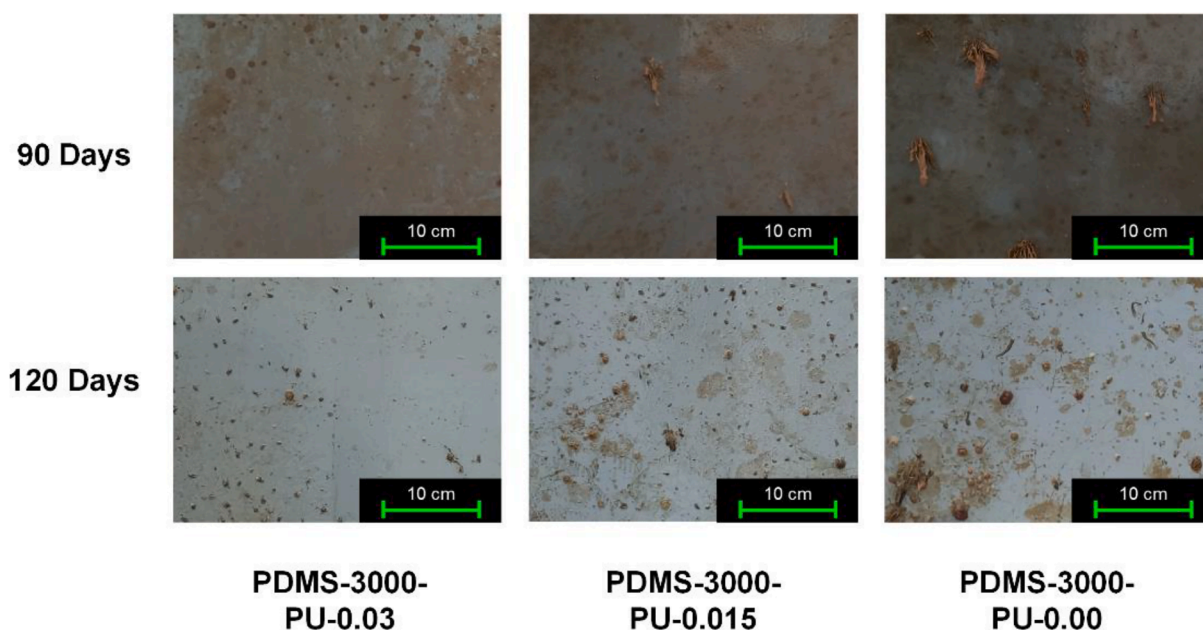


Fig. 5. Images of the tested panels coated with PDMS-3000-PU-y after immersion in seawater for 90 days and 120 days.

formation of good adhesion to the substrate. The excellent antifouling performance could be attributed to the contact antibacterial properties and fluorescence antifouling and the strong synergistic coupling effect endowed decreased addition (less than 0.20 wt%, Table S1) of antifouling groups. Bacterial biofouling and algae resistance tests proved the remarkably synergistic effect of two mechanisms during the settlement of the first and second stages of fouling. The PDMS-based polyurethane coating containing 7-amino-4-methylcoumarin based on the synergistic antifouling strategy is thus highly promising for an even broader range of protective applications like anti-biofouling, anti-corrosion, anti-icing, flame-resistance, self-cleaning as well as intelligent fluorescent response topcoat.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2020.127676>.

References

- [1] A.G. Nurioglu, A.C.C. Esteves, G. de With, Non-toxic, non-biocide-release antifouling coatings based on molecular structure design for marine applications, *J. Mater. Chem. B* 3 (2015) 6547–6570.
- [2] U. Eduok, O. Faye, J. Szpunar, Recent developments and applications of protective silicone coatings: A review of PDMS functional materials, *Prog. Org. Coat.* 111 (2017) 124–163.
- [3] J.A. Callow, M.E. Callow, Trends in the development of environmentally friendly fouling-resistant marine coatings, *Nat. Commun.* 2 (2011).
- [4] P. Hu, Q. Xie, C. Ma, G. Zhang, Silicone-Based Fouling-Release Coatings for Marine Antifouling, *Langmuir* 36 (2020) 2170–2183.
- [5] A.K. Leonardi, C.K. Ober, Polymer-Based Marine Antifouling and Fouling Release Surfaces: Strategies for Synthesis and Modification, in: J.M. Prausnitz (Ed.) *Annu Rev Chem Biomol* 2019, pp. 241–264.
- [6] H. Guo, J. Yang, W. Zhao, T. Xu, C. Lin, J. Zhang, L. Zhang, Direct formation of amphiphilic crosslinked networks based on PVP as a marine anti-biofouling coating, *Chem. Eng. J.* 374 (2019) 1353–1363.
- [7] M.S. Selim, M.A. Shenashen, S.A. El-Safty, S.A. Higazy, M.M. Selim, H. Isago, A. Elmarakbi, Recent progress in marine foul-release polymeric nanocomposite coatings, *Prog Mater Sci* 87 (2017) 1–32.
- [8] X. Gou, Z. Guo, Surface topographies of biomimetic superamphiphobic materials: design criteria, fabrication and performance, *Adv Colloid Interface Sci* 269 (2019) 87–121.
- [9] A. Ekin, D.C. Webster, Combinatorial and high-throughput screening of the effect of siloxane composition on the surface properties of crosslinked siloxane-polyurethane coatings, *J Comb Chem* 9 (2007) 178–188.
- [10] S. Sommer, A. Ekin, D.C. Webster, S.J. Stafslien, J. Daniels, L.J. VanderWal, S.E. M. Thompson, M.E. Callow, J.A. Callow, A preliminary study on the properties and fouling-release performance of siloxane-polyurethane coatings prepared from poly (dimethylsiloxane) (PDMS) macromers, *Biofouling* 26 (2010) 961–972.
- [11] R. Holland, T.M. Dugdale, R. Wetherbee, A.B. Brennan, J.A. Finlay, J.A. Callow, M. E. Callow, Adhesion and motility of fouling diatoms on a silicone elastomer, *Biofouling* 20 (2004) 323–329.
- [12] S. Krishnan, C.J. Weinman, C.K. Ober, Advances in polymers for anti-biofouling surfaces, *J. Mater. Chem.* 18 (2008).
- [13] Q. Ye, B. He, Y. Zhang, J. Zhang, S. Liu, F. Zhou, Grafting Robust Thick Zwitterionic Polymer Brushes via Subsurface-Initiated Ring-Opening Metathesis Polymerization for Antimicrobial and Anti-Biofouling, *ACS Appl. Mater. Interfaces* 11 (2019) 39171–39178.
- [14] M. Oliva, E. Martinelli, G. Galli, C. Pretti, PDMS-based films containing surface-active amphiphilic block copolymers to combat fouling from barnacles B. amphitrite and B. improvisus, *Polymer* 108 (2017) 476–482.
- [15] Q. Xie, C. Ma, C. Liu, J. Ma, G. Zhang, Poly(dimethylsiloxane)-Based Polyurethane with Chemically Attached Antifoulants for Durable Marine Antibiofouling, *ACS Appl. Mater. Interfaces* 7 (2015) 21030–21037.
- [16] J. Thomas, S.B. Choi, R. Fjeldheim, P. Boudjouk, Silicones containing pendant biocides for antifouling coatings, *Biofouling* 20 (2004) 227–236.
- [17] E.R. Silva, O. Ferreira, P.A. Ramalho, N.F. Azevedo, R. Bayon, A. Igartua, J. C. Bordado, M.J. Calhorda, Eco-friendly non-biocide-release coatings for marine biofouling prevention, *Sci. Total Environ.* 650 (2019) 2499–2511.
- [18] A. Ali, M.I. Jamil, J. Jiang, M. Shoaib, B.U. Amin, S. Luo, X. Zhan, F. Chen, Q. Zhang, An overview of controlled-biocide-release coating based on polymer resin for marine antifouling applications, *J. Polym. Res.* 27 (2020).
- [19] G. Achar, S. C. R, S.A. Patil, J.G. Malecki, S. Budagumpi, Coumarin-substituted 1,2,4-triazole-derived silver(i) and gold(i) complexes: synthesis, characterization and anticancer studies, *New Journal of Chemistry* 43 (2019) 1216–1229.
- [20] M. Tasior, D. Kim, S. Singha, M. Krzeszewski, K.H. Ahn, D.T. Gryko, π -Expanded coumarins: synthesis, optical properties and applications, *J. Mater. Chem. C* 3 (2015) 1421–1446.
- [21] H. Wei, J. Ruan, X. Zhang, Coumarin–chalcone hybrids: promising agents with diverse pharmacological properties, *RSC Adv.* 6 (2016) 10846–10860.
- [22] M.M. Liu, X.Y. Chen, Y.Q. Huang, P. Feng, Y.L. Guo, G. Yang, Y. Chen, Hybrids of phenylsulfonylfuroxan and coumarin as potent antitumor agents, *J Med Chem* 57 (2014) 9343–9356.

- [23] X.M. Peng, G.L. Damu, C. Zhou, Current developments of coumarin compounds in medicinal chemistry, *Curr Pharm Des* 19 (2013) 3884–3930.
- [24] P. Bhaskar, N.B. Bhosle, Microbial extracellular polymeric substances in marine biogeochemical processes, *Curr. Sci.* (2005) 45–53.
- [25] L. Cui, C.F. Ji, Z.X. Peng, L. Zhong, C.H. Zhou, L.L. Yan, S. Qu, S.P. Zhang, C. S. Huang, X.H. Qian, Y.F. Xu, Unique Tr-Output Optical Probe for Specific and Ultrasensitive Detection of Hydrazine, *Anal Chem* 86 (2014) 4611–4617.
- [26] J.T. Kirk, *Light and photosynthesis in aquatic ecosystems*, Cambridge university press 1994.
- [27] N.G. Jerlov, Ultra-Violet Radiation in the Sea, *Nature* 166 (1950) 111–112.
- [28] S. Cao, J. Wang, D. Chen, Influence of illumination on settlement of diatom *Navicula* sp, *Microb Ecol* 62 (2011) 931–940.
- [29] S. Cao, J.D. Wang, Y. Zhang, D.R. Chen, The effectiveness of an antifouling compound coating based on a silicone elastomer and colored phosphor powder against *Navicula* species diatom, *J. Coat. Technol. Res.* 10 (2013) 397–406.
- [30] H. Jin, W. Bing, E. Jin, L. Tian, Y. Jiang, Bioinspired PDMS–Phosphor–Silicone Rubber Sandwich-Structure Coatings for Combating Biofouling, *Adv. Mater. Interfaces* 7 (2020).
- [31] A.L. Friedman, R.S. Alberte, Biogenesis and light regulation of the major light harvesting chlorophyll-protein of diatoms, *Plant Physiol* 80 (1986) 43–51.
- [32] S.A. Cohn, S. Dunbar, R. Ragland, J. Schulze, A. Suchar, J. Weiss, A. Wolske, Analysis of light quality and assemblage composition on diatom motility and accumulation rate, *Diatom Res.* 31 (2016) 173–184.
- [33] A. Zielińska, E. Kowalska, J.W. Sobczak, I. Łacka, M. Gazda, B. Ohtani, J. Hupka, A. Zaleska, Silver-doped TiO₂ prepared by microemulsion method: Surface properties, bio- and photoactivity, *Sep. Purif. Technol.* 72 (2010) 309–318.
- [34] Y. Ding, B. Huang, H. Ou, F. Da Ros, C. Peucheret, Polarization diversity DPSK demodulator on the silicon-on-insulator platform with simple fabrication, *Opt Express* 21 (2013) 7828–7834.
- [35] Y.K. Kim, S.M. Sharker, I. In, S.Y. Park, Surface coated fluorescent carbon nanoparticles/TiO₂ as visible-light sensitive photocatalytic complexes for antifouling activity, *Carbon* 103 (2016) 412–420.
- [36] Y. Fu, J. Jiang, Q. Zhang, X. Zhan, F. Chen, Robust liquid-repellent coatings based on polymer nanoparticles with excellent self-cleaning and antibacterial performances, *J Mater Chem A* 5 (2017) 275–284.
- [37] T. Gurunathan, J.S. Chung, Physicochemical Properties of Amino–Silane-Terminated Vegetable Oil-Based Waterborne Polyurethane Nanocomposites, *ACS Sustainable Chem. Eng.* 4 (2016) 4645–4653.
- [38] T.P. Galhenage, D.C. Webster, A.M.S. Moreira, R.J. Burgett, S.J. Stafslie, L. Vanderwal, J.A. Finlay, S.C. Franco, A.S. Clare, Poly(ethylene) glycol-modified, amphiphilic, siloxane-polyurethane coatings and their performance as fouling-release surfaces, *J. Coat. Technol. Res.* 14 (2017) 307–322.
- [39] J. Jiang, Y. Fu, Q. Zhang, X. Zhan, F. Chen, Novel amphiphilic poly (dimethylsiloxane) based polyurethane networks tethered with carboxybetaine and their combined antibacterial and anti-adhesive property, *Appl. Surf. Sci.* 412 (2017) 1–9.
- [40] R.F. Brady Jr, Properties which influence marine fouling resistance in polymers containing silicon and fluorine, *Prog. Org. Coat.* 35 (1999) 31–35.
- [41] R.E. Baier, Surface behaviour of biomaterials: the theta surface for biocompatibility, *J. Mater. Sci. - Mater. Med.* 17 (2006) 1057.
- [42] B. Jin, G. Zhang, J. Lian, Q. Zhang, X. Zhan, F. Chen, Boron nitride nanosheet embedded bio-inspired wet adhesives with switchable adhesion and oxidation resistance, *J Mater Chem A* 7 (2019) 12266–12275.
- [43] F. Liu, F. Sun, Q. Pan, Highly compressible and stretchable superhydrophobic coating inspired by bio-adhesion of marine mussels, *J. Mater. Chem. A* 2 (2014) 11365–11371.
- [44] S. Gafner, C. Bergeron, J.R. Villinski, M. Godejohann, P. Kessler, J.H. Cardellina, D. Ferreira, K. Feghali, D. Grenier, Isoflavonoids and coumarins from *Glycyrrhiza uralensis*: antibacterial activity against oral pathogens and conversion of isoflavans into isoflavan-quinones during purification, *J Nat Prod* 74 (2011) 2514–2519.